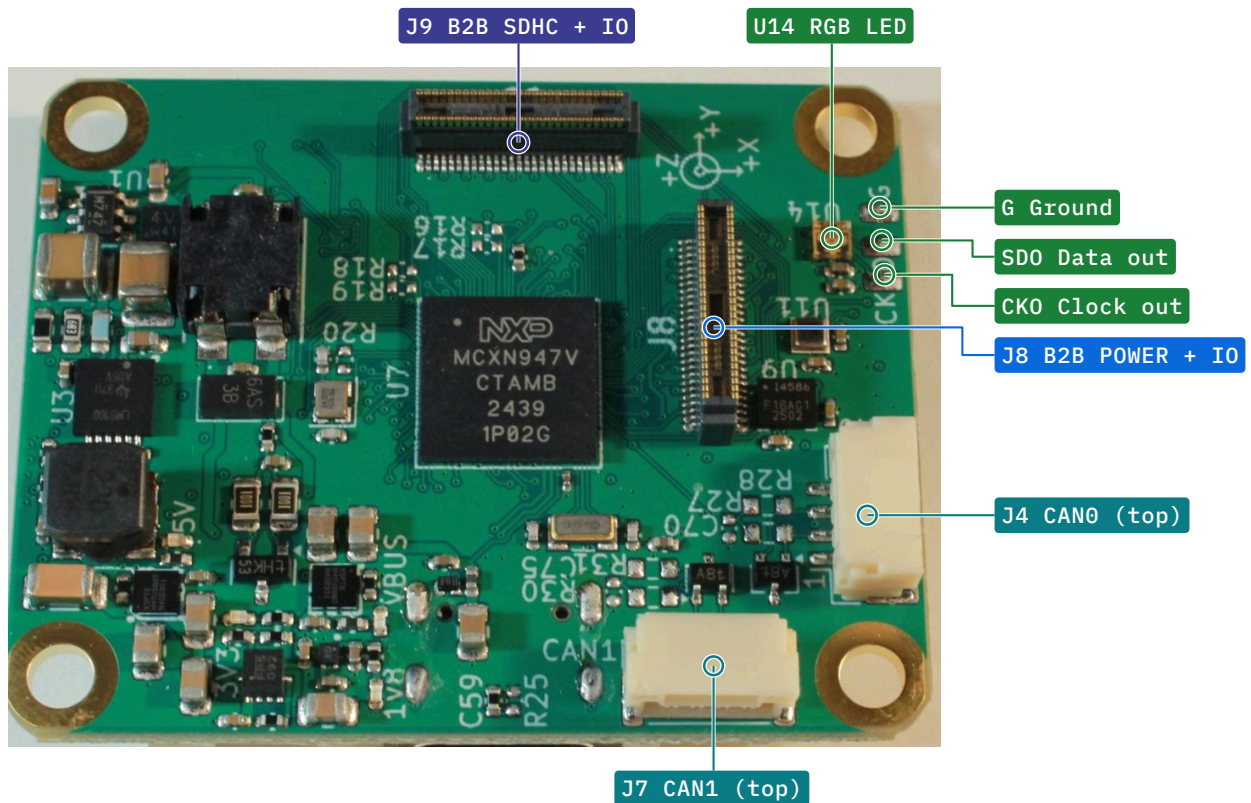


MR-MCXN-T1 Hub hardware reference

Connectors and pinouts of the NXP MR-MCXN-T1 Hub, an MCXN947 base board with a board-to-board interface for add-on boards. All data comes from the KiCad design in the [mr_mcxn_t1_hub repository](#).

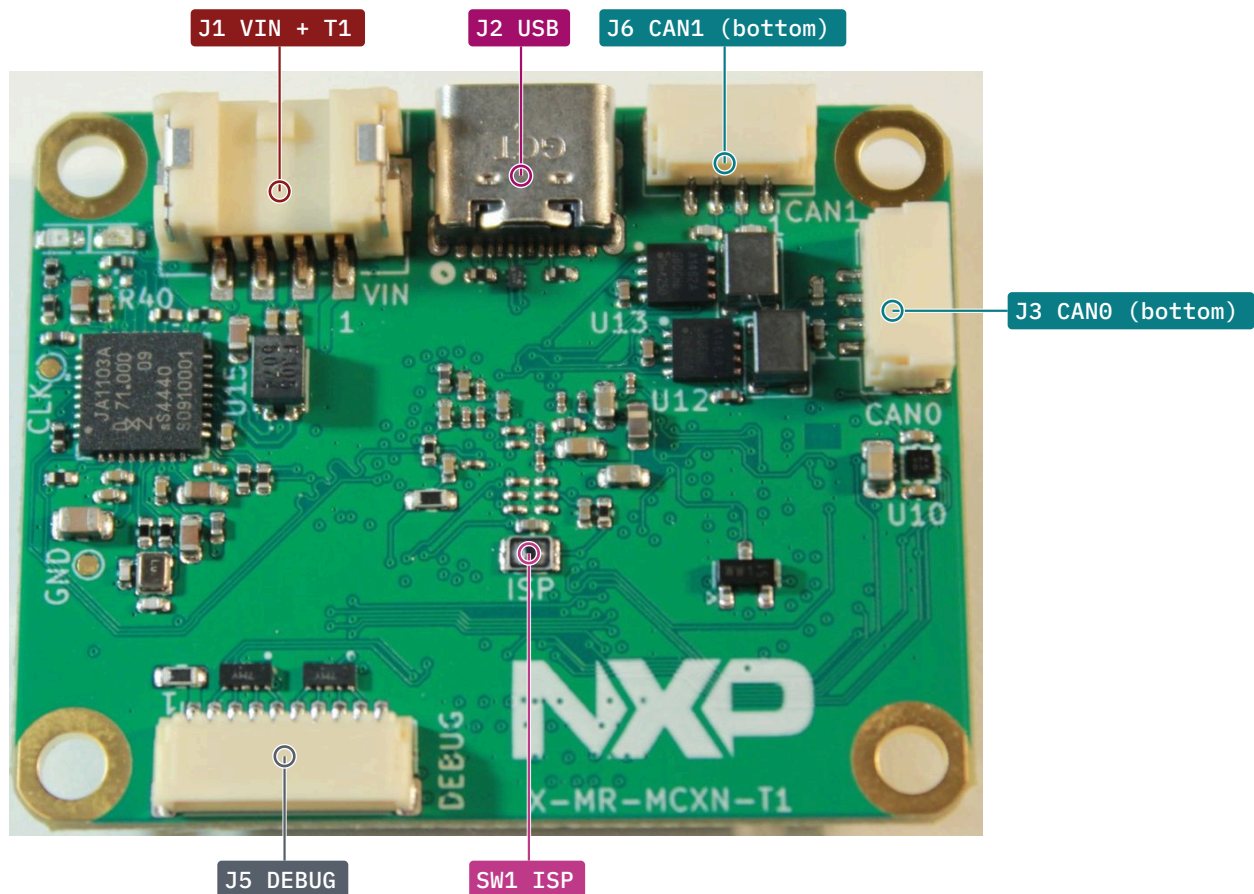
Each card shows, per pin, the **function**, the **net name in the schematic** and the **MCU pad** the signal ends up on. Parts in between, such as the CAN transceivers and the Ethernet PHY, are shown too. Pin 1 is the square pad.

Board overview



Bottom side: **J1 VIN + T1** · **J2 USB** · **J5 DEBUG** · **J13 RTC_BAT** · **SW1 ISP**.

Bottom side, with the power and Ethernet connector, USB, debug port and the bottom CAN connectors:



New in the current revision, bottom right corner (the photo shows the old revision): [J13 RTC_BAT](#).

Connector index

Ref	Function	Connector	Details
Hub J1 VIN + T1	Battery input 3S to 12S and 100BASE-T1 Ethernet on one cable	Molex Micro-Lock Plus 1x4 (505578-0421), 1.25 mm, side entry	
Hub J2 USB	USB 2.0 high-speed device port	USB Type-C receptacle (USB4105-GF-A)	On USB1 of the MCXN947 (high speed). The board is always a USB device (CC pull-downs). VBUS powers the board through load switch U2 when no battery is connected. With SW1 held at power up the boot ROM listens here for ISP programming.
Hub J3 CAN0	CAN FD bus 0 (TJA1462), two connectors in parallel	JST-GH 1x4, J3 on the bottom side and J4 on the top side	
Hub J5 DEBUG	SWD, JTAG and console, DS-009 debug full layout	JST-SH 1x10	
Hub J6 CAN1	CAN FD bus 1 (TJA1462), two connectors in parallel	JST-GH 1x4, J6 on the bottom side and J7 on the top side	

Ref	Function	Connector	Details
Hub J8 B2B POWER + IO	Add-on board connector 1: power rails and FlexComm, I3C, analog and trace pins	Hirose DF40 50-pin header, 0.4 mm pitch, 3.0 mm stack (DF40TC(3.0)-50DP-0.4V)	
Hub J9 B2B SDHC + IO	Add-on board connector 2: SDHC or FlexComm 9, SmartDMA or FlexComm 6/7, and more FlexComm pins	Hirose DF40 50-pin header, 0.4 mm pitch, 3.0 mm stack (DF40TC(3.0)-50DP-0.4V)	
Hub J13 RTC_BAT	Backup battery for the RTC	JST-SH 1x2, side entry	
Hub RGB LED CHAIN	Output of the on-board RGB LED, solder pads	Three pads marked CKO, SDO and G next to U14, 5 V logic	

Power, Ethernet and USB

Hub J1 VIN + T1

J1 VIN + T1

Molex Micro-Lock Plus 1x4 (505578-0421), 1.25 mm, side entry



Pin 1 takes the battery, 3S to 12S (about 9 to 50 V), 10 W budget. An ideal diode (LM74700) blocks reverse polarity, a buck converter makes 5 V from it, and 3.3 V and 1.8 V come from the 5 V rail. The 5 V rail switches on above about 8 V. Pins 2 and 3 are the 100BASE-T1 pair, one twisted pair to the TJA1103 PHY. The PHY is the link slave (R41 fitted); remove R41 to make it the master. The other end of the T1 link must be the master. Mating parts from the schematic: socket 505570-0401, crimp 505572-1100, pre-crimped wires 217500-1125 (black) and 217500-2125 (red).

Caution

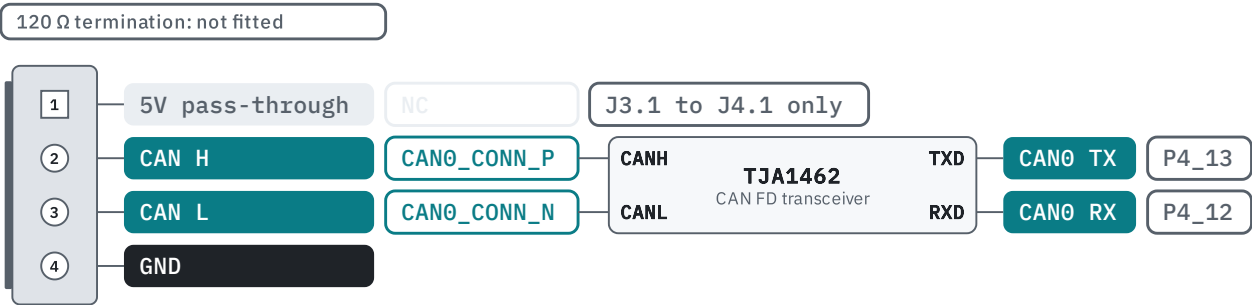
Pin 1 is the only power input besides USB. Keep the battery above 3S, the board does not start below about 8 V.

CAN

Hub J3 CAN0

J3 CAN0

JST-GH 1x4, J3 on the bottom side and J4 on the top side · DS-009 CAN



J3 (bottom) and J4 (top) carry the same bus, so one cable can come in and another go out. The transceiver runs from 5 V and talks to the chip at 3.3 V. Drive P4_14 (CAN0_STB) low to enable it. The split termination (R27, R28 and C70) is not fitted, so the bus must be terminated elsewhere, or fit the parts if the hub is one end of the bus.

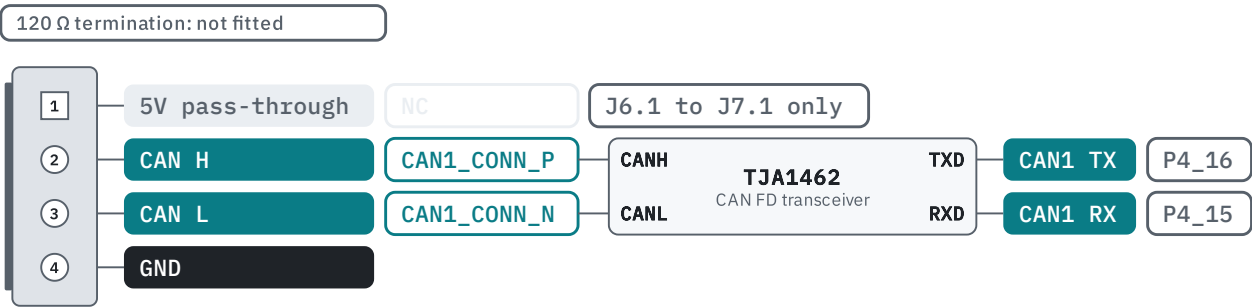
Note

Pin 1 is not powered. It is only linked between J3 and J4, so a 5 V supply on one cable passes to the other.

Hub J6 CAN1

J6 CAN1

JST-GH 1x4, J6 on the bottom side and J7 on the top side · DS-009 CAN



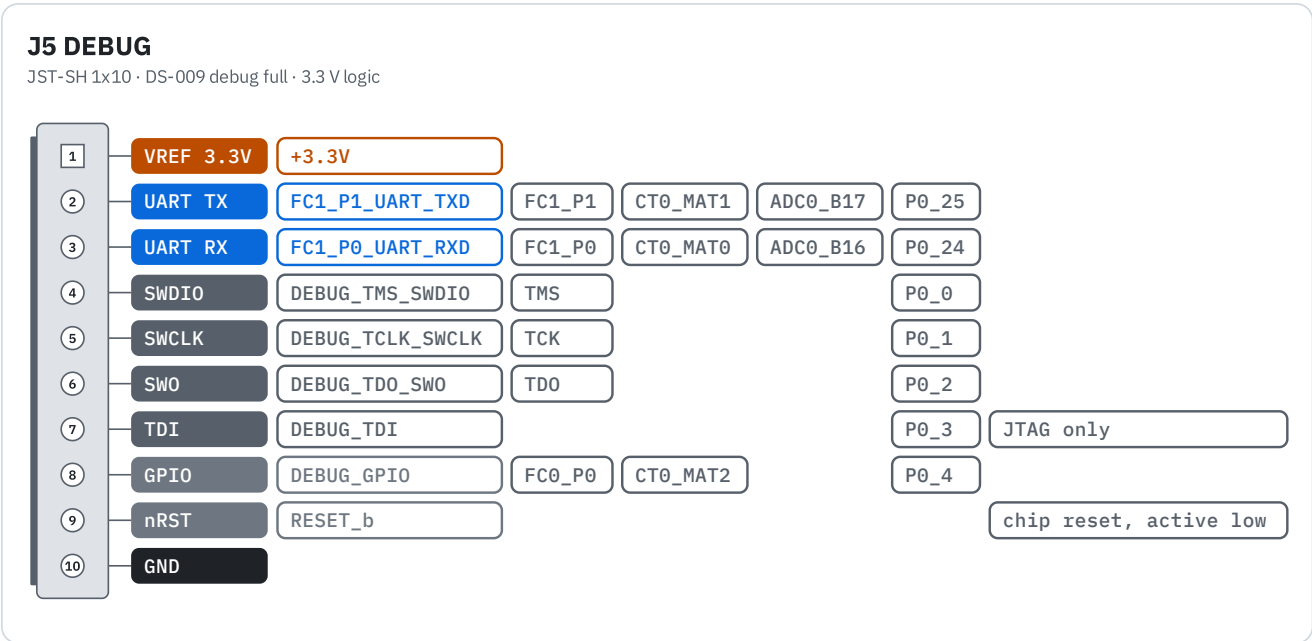
Same circuit as CAN0. J6 (bottom) and J7 (top) carry the same bus. Drive P4_17 (CAN1_STB) low to enable the transceiver. The split termination (R30, R31 and C75) is not fitted.

Note

Pin 1 is not powered. It is only linked between J6 and J7, so a 5 V supply on one cable passes to the other.

Debug

Hub J5 DEBUG



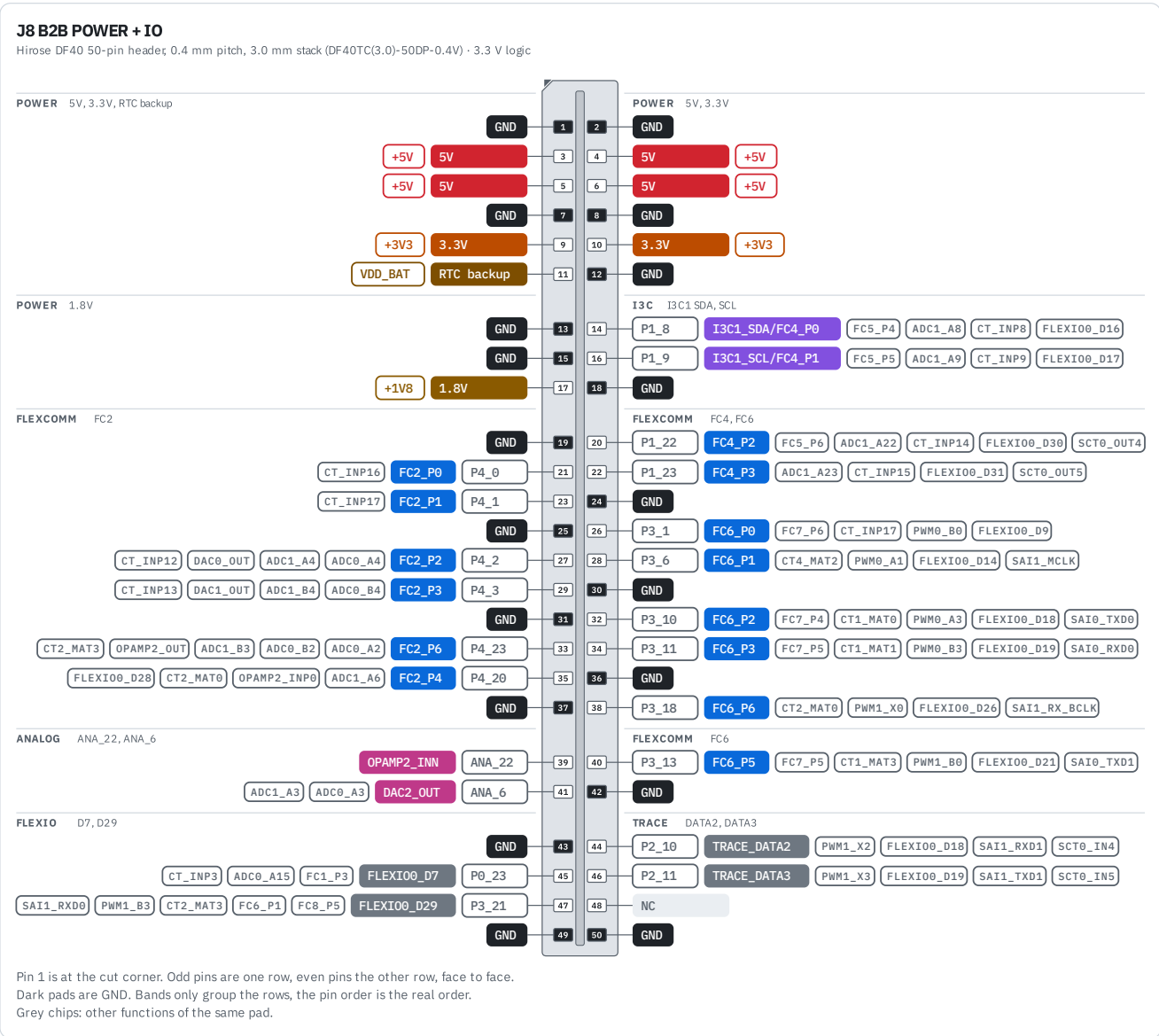
Pin 1 is a 3.3 V reference output for the probe. Pins 2 and 3 are FlexComm 1 as a UART console. Pins 4 to 7 are the SWD and JTAG lines of the MCXN947, pin 8 is a spare GPIO for the probe, pin 9 is the chip reset, pulled up on the board.

Warning

Pin 1 is a reference output, not a supply input. Never power the board from it.

Board-to-board interface

Hub J8 B2B POWER + IO

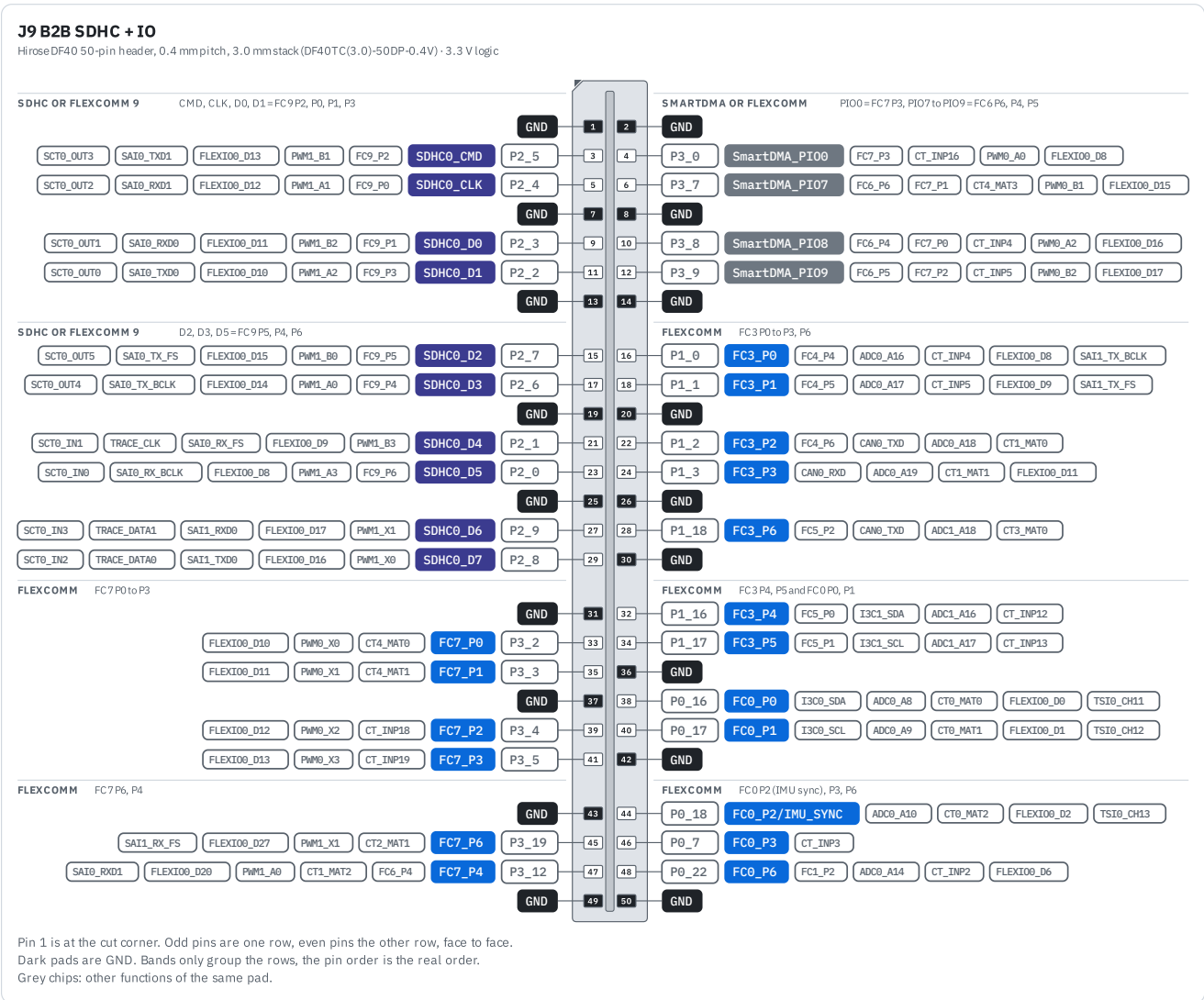


Together with J9 this is the interface for the add-on boards (camera, audio, SDIO, mikroBUS, display, ADC, IO). Odd pins are one row, even pins the other, pin 1 at the cut corner. Limits from the schematic: 300 mA per pin, 5 V rail 1 A, 3.3 V rail 500 mA, 1.8 V rail 200 mA. Pin 11 is the RTC backup supply, shared with J13. The FlexComm blocks (FC2, FC4, FC6) can each be a UART, SPI or I2C. The grey chips list the other functions of each pad.

Tip

The add-on board designs are in the `spinali_mcxn_t1_shields` repository, linked from the main README.

Hub J9 B2B SDHC + IO



Second add-on connector. No power pins here, all supplies are on J8. Every signal pin has more than one job, that is the point of this interface. The SDHC0 pins carry an 8-bit SD or eMMC interface, but the same pads are FlexComm 9 (P0 to P6), so an add-on board can use them as one more UART, SPI or I2C instead. The SmartDMA pins are a parallel camera port and double as FlexComm 6 and 7 lines or PWM0 outputs. FC0, FC3 and FC7 can each be a UART, SPI or I2C. Pin 44 is also the IMU sync input. The grey chips list the other functions of each pad.

LED chain and RTC battery

Hub RGB LED CHAIN



The MCU drives one SK9822 RGB LED (U14, APA102 compatible) from FlexIO pins P0_8 (data) and P0_9 (clock). CKO and SDO are the clock and data outputs of that LED, so more LEDs of the same type can be chained from here. The LED runs from 5 V, so its outputs are 5 V.

Hub J13 RTC_BAT

J13 RTC_BAT

JST-SH 1x2, side entry



Keeps the real-time clock of the MCXN947 running without main power. Fits the RPi RTC battery (ML2020 with a 2-pin JST-SH plug). The same supply is on J8 pin 11 for the add-on board. The board charges the cell from 3.3 V through D15 and a 75 Ω resistor.

Warning

The board trickle-charges this cell. Use a rechargeable ML2020 type, never a CR2032 or other non-rechargeable cell.

Hub other connectors

Ref	Function	Connector	Notes
J2 USB	USB 2.0 high-speed device port	USB Type-C receptacle (USB4105-GF-A)	On USB1 of the MCXN947 (high speed). The board is always a USB device (CC pull-downs). VBUS powers the board through load switch U2 when no battery is connected. With SW1 held at power up the boot ROM listens here for ISP programming.

Buttons and switches

Ref	Board	Function	Type	Notes
SW1 ISP	Hub	ISP boot button	Tactile switch, bottom side	Hold it while power comes up to start the boot ROM in ISP mode (P0_6 ISPMODE_N pulled low), for programming over USB or the debug UART. Normal boot otherwise.

On-board devices

How the sensors, the PHY and the LED are wired to the MCXN947. Useful for a device tree or board file.

Device	Bus	MCU pins	Notes
ICM-45686 IMU (U9)	SPI on FlexComm 8	SCK P3_15, MOSI P3_14, MISO P3_16, CS P3_17, INT1 P3_22, FSYNC/INT2 P0_18	P0_18 is also on J9 pin 44 as IMU sync.
BMM350 magnetometer (U10)	I3C0 (or I2C)	SDA P0_20, SCL P0_21, INT P0_19	Address 0x14. Runs from 1.8 V.
BMP581 barometer (U11)	I3C0 (or I2C)	SDA P0_20, SCL P0_21, INT P0_11	Address 0x47 (SDO pulled up, R48 not fitted).

Device	Bus	MCU pins	Notes
TJA1103 100BASE-T1 PHY (U15)	RMII on ENET0	TXD0 P1_6, TXD1 P1_7, TXEN P1_5, TXER P1_10, RXD0 P1_14, RXD1 P1_15, RXDV P1_13, RXER P1_12, REFCLK P1_4, MDC P1_20, MDIO P1_21, INT P1_19, RESET P1_11, WAKE P0_31	Reverse RMII: the PHY supplies the 50 MHz reference clock. PHY address 18. Link slave, because R41 is fitted; remove R41 for master. The two LEDs next to J1 are driven by the PHY GPIO0 and GPIO1.
TJA1462 CAN FD transceivers (U12, U13)	FlexCAN0, FlexCAN1	CAN0 TX P4_13, RX P4_12, STB P4_14; CAN1 TX P4_16, RX P4_15, STB P4_17	Drive STB low to enable a transceiver.
SK9822 RGB LED (U14)	FlexIO	data P0_8, clock P0_9	APA102 compatible, 5 V. Chain output on the CKO, SDO and G pads.
ISP button (SW1)	GPIO	P0_6	ISPMODE_N, boot ROM ISP mode when held at power up.

Test pads on the bottom side: CLK is the PHY reference clock (P1_4), GND is ground.

Notes

- The JST-GH and JST-SH connectors follow the Dronecode [DS-009 connector standard](#) (CAN and debug full), so standard cables fit. The CAN pin 1 is not powered on this board, and the CAN termination resistors are not fitted.
- The board runs from the battery on J1 or from USB. 5 V, 3.3 V and 1.8 V for an add-on board come from J8.
- The photos show the old revision, marked X-MR-MCXN-T1. The RTC battery connector J13 was added in the current revision, in the bottom right corner of the bottom side.

Warning

Never feed power into the 5 V, 3.3 V or 1.8 V pins of J8, or into pin 1 of the debug port. Power the board through J1 or USB only.

Downloads

This reference as an [A4 PDF](#), and all cards on one A4 landscape sheet: [cheat sheet PDF](#).

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